ERMYAS MISIKER

Full-Stack & Software Developer • AI Systems Engineer

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PROFESSIONAL SUMMARY

Self-taught software engineer started from age 15. Strong foundation in backend systems architecture, API design, on-device AI models, and cross-platform application development. Proven track record delivering complete production software systems with **Java / Spring Boot 3**, **Python (FastAPI)**, **Node.js / Express**, **React**, and **Flutter**. Architect of client-side Amharic AI meeting systems (Weg AI) and offline on-device speech translation engines (LISAN AI). Passionate about enterprise backend engineering, AI automation, and scalable systems.

TECHNICAL SKILLS

Languages:	Java (17+), Python, TypeScript, JavaScript, Dart, Kotlin, C++, SQL, HTML5/CSS3
Backend Frameworks:	Spring Boot 3 (Spring Security 6, Spring Data JPA, Hibernate), FastAPI, Node.js, Express.js, Django
Frontend & Mobile:	React, Next.js, Vite, Flutter, Tailwind CSS, Responsive Web Design, Android (Kotlin), Chrome MV3
Databases & Cloud:	PostgreSQL, Supabase (Auth, RLS, Edge Functions, Realtime), SQLite, H2 Database, Docker
APIs & Security:	RESTful APIs, OpenAPI 3 / Swagger UI, JWT Authentication & RBAC, WebSockets / Socket.io
AI & Speech / Automation:	On-Device Speech AI (VITS MMS-TTS, Quantized NLLB-200, C++ FFI zero-copy, Hohe ASR), LLM APIs (Gemini 2.5 Flash, Groq LLaMA-3.3, OpenAI), Prompt Engineering, RAG Pipelines, Tesseract OCR, Telegram Bot API
Tools & Platforms:	Git, GitHub, Maven, Docker, Postman, Vercel, Render, Linux / Bash

FEATURED SOFTWARE PROJECTS & SYSTEMS

Weg (ወግ) AI — Real-Time Meeting Intelligence Assistant — **Chrome MV3 • Android Kotlin • FastAPI • Gemini 2.5 Flash • Hohe ASR**

- **Architected client-side meeting assistant** across Chrome MV3 and Kotlin Android app, capturing tab/system audio directly without bot participants.
- **Engineered real-time audio chunk streaming** (5-second sequential buffers) with Gemini 2.5 Flash and Hohe ASR for structured bilingual note generation.
- **Built high-throughput FastAPI backend** services for asynchronous processing, transcript caching, and action item synthesis.

LISAN (ልሳን) AI — Offline Speech-to-Speech Translation Mobile App — **Flutter • C++ FFI • ONNX Runtime • VITS MMS-TTS • NLLB-200**

- **Engineered offline speech-to-speech translation** across 4 Ethiopian languages (Amharic, Afaan Oromoo, Tigrinya, Somali, English) under strict **350MB RAM limit**.
- **Implemented zero-copy C++ FFI decoding** for on-device VITS MMS-TTS and 8-bit quantized NLLB-200 via ONNX Runtime without network dependence.
- **Built crowdsourced validation ecosystem** & gamified Telegram bot with real-time XP leaderboards, verifying over **373K+ bitext pairs**.

Healthcare Appointment Management API — **Java 17 • Spring Boot 3 • JPA • PostgreSQL**

- **Architected enterprise REST API** for hospital scheduling, patient records, and doctor management using layered architecture.
- **Implemented stateless JWT authentication** and role-based access control (Admin, Doctor, Patient) using Spring Security 6.
- **Optimized relational persistence** with Spring Data JPA & Hibernate, incorporating custom DTOs, bean validation, and pagination.
- **Documented & tested endpoints** with OpenAPI 3 / Swagger UI, integrating MockMvc and JUnit 5 test suites.

USDENT Appeal Builder — Denial Automation SaaS — **Python • FastAPI • Next.js • Groq (LLaMA-3.3)**

- **Automated dental insurance EOB appeals** by extracting denial codes and line items via Tesseract OCR document parsing.
- **Integrated Groq API (LLaMA-3.3-70B)** to evaluate clinical denial rules and generate structured, payer-specific appeal narratives.
- **Engineered automated PDF mapping** to populate standardized ADA 2024 dental claim forms and bundle submission packets.

LINC — AI-Powered Service Discovery Platform — **Node.js • Express • React • Supabase • Gemini API**

- **Engineered monorepo marketplace** connecting clients and verified providers with a clean 4-layer backend architecture.
- **Designed 20-table PostgreSQL schema** on Supabase for provider profiles, multi-tier services, verification, and bookings.
- **Integrated Google Gemini AI pipeline** for natural language intent parsing and provider match scoring (budget, rating, location).
- **Built real-time communication layer** using Socket.io for direct chat messaging and instantaneous push notifications.

E.mslene (Sunday School Management Platform) — **Flutter • Dart • React • Supabase PostgreSQL • RBAC**

- **Built multi-role educational system** serving students, teachers, and administrators across mobile and web clients.
- **Developed cross-platform mobile client** in Flutter for digital registration, live attendance tracking, and grade disclosure.
- **Created React + Vite admin dashboard** with granular Role-Based Access Control (RBAC) and Row-Level Security (RLS).

Space For ET & AI Telegram Bot Suite — **Flutter • Dart • Python • Supabase • Telegram Bot API**

- **Published Flutter mobile application** providing live rocket launch alerts, space news, and localized Ethiopian stargazing data.
- **Built interactive Telegram AI tutor** with persistent user memory on Supabase, plus automated daily space news delivery bots.

EDUCATION & TRAINING

BSc in Software Engineering — Addis Ababa University (AAU) — **Expected 2030 (Enrolled 2025)**

Core Focus: Software Architecture, Data Structures & Algorithms, OOP (Java/Python), Database Systems, System Design.

Software Engineering & Cybersecurity Trainee (5th Round) — INSA — **Addis Ababa, Ethiopia**

Information Network Security Administration: Network Security Fundamentals, Secure Coding Practices, Threat Modeling, Enterprise Systems.